

Performance Evaluation of Machine Learning Models for Phishing Website Detection

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Abstract

Phishing attacks are one of the most common cybersecurity threats, aiming to deceive users into revealing sensitive information such as passwords and financial details. Detecting phishing websites manually is difficult due to the increasing sophistication of attacks. This paper presents a comparative analysis of machine learning models for phishing website detection using a publicly available dataset. Three models—K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Random Forest—are implemented and evaluated. Experimental results show that the Random Forest model achieves the highest accuracy of 96.65%, outperforming other models. The study demonstrates the effectiveness of ensemble learning techniques in cybersecurity applications

1. Introduction:

Phishing is a cyber attack in which
Introduction

attackers impersonate legitimate websites to steal user information. With the rapid growth of internet usage, phishing attacks have become more frequent and sophisticated. Traditional rule-based detection systems are no longer sufficient to handle modern phishing techniques. Machine Learning (ML) provides an automated approach to detect phishing websites by analyzing patterns in data. This paper focuses on implementing and comparing different ML models to identify the most effective method for phishing detection.

2. Literature Review

Previous research has explored various machine learning techniques for phishing detection. Studies have shown that models such as Decision Trees, Support Vector Machines, and Neural Networks can effectively classify phishing websites.

The dataset used in this study is based on the work by R. Mohammad et al. (2012), which

includes features extracted from phishing and legitimate websites. These features capture URL characteristics, domain properties, and webpage behavior.

3. Methodology

3.1 Dataset

The dataset contains 11,055 instances and 30 features representing different characteristics of websites. The target variable indicates whether a website is phishing or legitimate.

3.2 Data Preprocessing

- The dataset was cleaned by removing irrelevant columns such as the index.
- The target variable was converted from (-1, 1) to (0, 1).
- Features (X) and target (y) were separated.
- The dataset was split into training (80%) and testing (20%) sets.

3.3 Machine Learning Models

The following models were implemented:

- K-Nearest Neighbors (KNN): Classifies data based on nearest neighbors.

- Support Vector Machine (SVM): Uses hyperplanes for classification. Feature scaling was applied to improve performance.
- Random Forest: An ensemble method that combines multiple decision trees for better accuracy.

4.Results and Discussion

4.1 Model Performance

Model	Accuracy
KNN	93.66%
SVM	95.25%
Random Forest	96.65%

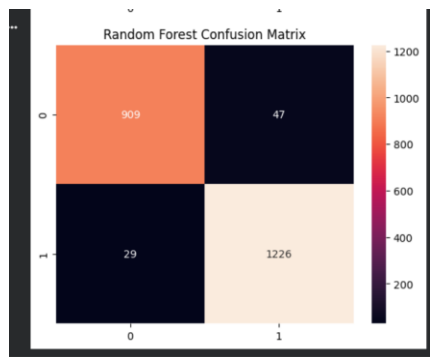
4.2 Classification Metrics

Random Forest achieved the highest precision, recall, and F1-score, indicating balanced and reliable performance. KNN and SVM also performed well but were slightly less accurate.

4.3 Confusion Matrix Analysis

The confusion matrix shows that Random Forest correctly classifies most phishing and legitimate websites

with minimal misclassification.



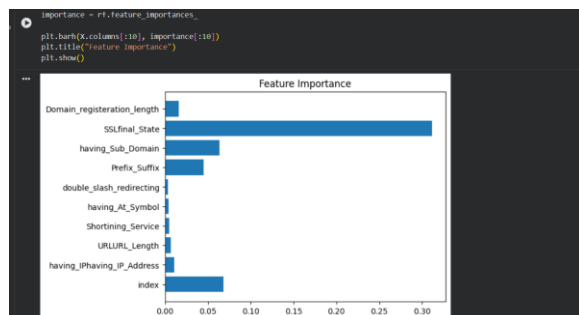
4.4 Graphical Analysis

- Accuracy comparison graph shows Random Forest as the best model.



ROC curve indicates strong classification performance.

Feature importance highlights key attributes such as SSL state, URL structure, and domain features.



5. Conclusion

This study compares three machine learning models for phishing website detection. Among them, the Random Forest model achieved the highest accuracy of 96.65%, making it the most effective model for this task.

Machine learning proves to be a powerful tool in cybersecurity, enabling automated and accurate detection of phishing attacks.

References:

1] R. Mohammad, F. Thabtah, and L. McCluskey,

“An assessment of features related to phishing websites using an automated technique,” International Conference for Internet Technology and Secured Transactions, 2012.